PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

CATTLE AI - INTELLIGENCE SYSTEM

A PROJECT REPORT

Submitted by

ASHWIN R - 20221CAI0042

SACHIN S - 20221CAI0026

HARSHIT B - 20221CAI0017

Under the guidance of,

Mr. JAI KUMAR B

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

(Artificial Intelligence and Machine Learning)

PRESIDENCY UNIVERSITY

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "Cattle AI - Intelligence System" is a bonafide work of "ASHWIN R (20221CAI0042), SACHIN S (20221CAI0026), HARSHIT B (20221CAI0017)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING (SPECIALIZATION DIVISION), ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING during 2025-26..

Dr. Jai Kumar B
Assistant Professor
Project Guide
PSCS
Presidency University

Ms. Suma N G
Assistant professor
Program Project Coordinator
PSCS
Presidency University

Dr. Sampath A K
Professor
School Project Coordinators
PSCS
Presidency University

Dr. Zafar Ali Khan N
Professor & Head of the Department
PSCS
Presidency University

Dr. Shakkeera L
Associate Dean
PSCS (Spl. Division)
Presidency University

S.No	Name of the Examiner	Signature	Date
1.	Jijina M T		25/4/26.
2.	Jasmine David D	Jasmine David	25/04/2026.

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SPECIALIZATION DIVISION)

DECLARATION

We, the students of the final year of B.Tech in COMPUTER SCIENCE ENGINEERING, (Spl. Division) at Presidency University, Bengaluru, named ASHWIN R, SACHIN S, and HARSHIT B, hereby declare that the project work titled **"CATTLE AI - INTELLIGENCE SYSTEM"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING (Spl. Division), during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any degree or diploma to any other institution.

ASHWIN R	20221CAI0042	
SACHIN S	20221CAI0026	
HARSHIT B	20221CAI0017	

PLACE: BENGALURU

DATE: 25-04-2026.

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ASHWIN R

SACHIN S

HARSHIT B

Abstract

Proper evaluation of cattle weight and breed is critical to the proper management of livestock and health surveillance. The old approach of using manual measurements and weighing scales is time-consuming, expensive and in many ways stressful to animals. Such difficulties are more pronounced in rural and semi-urban regions where the access to the appropriate equipment and expertise is scarce. Consequently, automated, non-invasive and cost-effective solutions are needed.

The CATTLE AI -Intelligence System is a deep learning-based model, which detects cattle, extracts features, classifies them by their breed, and estimates their weight based on images. Images are taken at various angles (side and rear) and processed by the system to extract important structural features of the animal. Cattle detection is performed with a YOLO-based model and then the segmentation is performed to isolate the animal and the background. The detection of key points and features is then utilized to extract useful anatomical measurements.

The system is constructed to be a modular pipeline composed of image preprocessing, detection, segmentation, feature extraction, and task-specific prediction modules. The classification of the breeds is done with the help of a CNN-based model and the estimation of the weight is done with the help of a hybrid approach of deep learning and regression.

It is compatible with real time inference and can be implemented on the commercial off the shelf (COTS) hardware, which is suitable to implement in order to perform livestock management.